# Hemant Kumar

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## Research Summary

My research lies at the intersection of robot learning, multimodal perception and control theory, with the goal of developing robots that can reason, adapt and act reliably in unstructured environments. To advance this vision, I work across foundation robotics models, and policy learning, connecting vision-language representations, action-conditioned reasoning, and control-theoretic principles to enable adaptive and generalizable robot behavior.

## Education

**University of Maryland, College Park** — Sept 2022 – Present
Ph.D. in Robotics — Advisor: Dr. Nikhil Chopra
Thesis Direction: Autonomous Disassembly of EVs (Electric Vehicles).
Adaptive Robotics · Computer Vision · Control Theory · Imitation Learning · Reinforcement Learning · VLAs · VLMs

**Indian Institute of Technology (IIT), Kharagpur** — Sept 2015 – May 2020
Bachelors (Honors) and Masters of Technology - Aerospace Engineering — Advisor: Dr. Chetan S. Mistry

## Patents

- **Hemant Kumar (Lead Inventor)**, Kaustubh Joshi, Nikhil Chopra,*"RACE: Real-Time Adaptive Camera Intrinsic Estimation",* (US Patent), Invention of the Year Award-Finalist, University of Maryland (2025).
- Piyush Goenka, **Hemant Kumar**, Kaustubh Joshi, Alexander Beyer, Tianchen Liu, Nikhil Chopra, *"UniScrew: Electronically Actuated Universal Electric Screwdriver Adapter for Any Robot Arm",* ( US-Patent).

## Research Publications

- **Hemant Kumar***, Piyush Goenka*, Nikhil Chopra, *"Toward Generalizable Robotic Disassembly: A Visuo-Tactile Pipeline for Screw Unfastening,"* Journal of Robotics and Autonomous Systems (**RAS 2026**). [Link]
- **Hemant Kumar**, Jamin Day, Sangeethaa Sivakumar, Nikhil Chopra, *"Do Multimodal Foundation Models Understand Assembly Procedures?,"* Workshop on Creativity in Assembly, International Conference on Intelligent Robots and Systems (**IROS 2026**).
- **Hemant Kumar**, Jamin Day, Sangeethaa Sivakumar, Nikhil Chopra, *"Benchmarking Vision-Language Models for Disassembly Tasks,"* AAAI Conference on Artificial Intelligence (**AAAI 2027**), under review.
- **Hemant Kumar**, Kaustubh Joshi, Nikhil Chopra, *"RACE: Real-Time Adaptive Camera-Intrinsic Estimation via Control Theory,"* IEEE Transactions on Robotics (**T-RO 2026**), under review.
- **Hemant Kumar**, Nikhil Chopra, *"Adaptive PTZ: Online Camera Self-Calibration under Dynamic Zoom and Motion,"* IEEE Robotics and Automation Letters (**RA-L 2026**), under review.
- **Hemant Kumar**, Piyush Goenka, Nikhil Chopra, *"DAWN: Hierarchical Diffusion Policies for Contact-Rich Robotic Unfastening,"* IEEE Robotics and Automation Letters (**RA-L 2026**), under review.
- **Hemant Kumar**, Nikhil Chopra, *"Generalizable Adaptive Camera Calibration for Zoom-Adaptive 3D Gaussian Splatting,"* IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2027**), under review.
- **Hemant Kumar**, Kaustubh Joshi, Nikhil Chopra, *"Online Adaptive Calibration for Nonlinear Wide-Angle Camera Models,"* IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2027**), under review.

## Selected Research Projects

- **DAWN: Hierarchical Diffusion Policies for Contact-Rich Robotic Unfastening**
  - A *framework* for long-horizon, contact-rich robotic unfastening, trained on imitation data and evaluated across **1,500+ physical trials** on a **Franka Emika robot**, demonstrating generalization to unseen fasteners and operating conditions.
- **ManualBench: Benchmarking Vision-Language Models for Assembly and Disassembly**
  - A multimodal benchmark comprising *68 real-world assembly manuals* and *136 assembly /disassembly task*

*graphs*, evaluating **20+ Vision-Language Models** on procedural reasoning and task planning, revealing limitations in *global dependency reasoning and parallel execution*.

- **Structure-Aware Learning for Procedural Reasoning and Planning** *(Ongoing)*
  - Developing *structure-aware learning methods* for multimodal foundation models to improve **long-horizon procedural reasoning**, and parallel task planning for complex robotic assembly and disassembly.
- **Dynamics-Adaptive Robot Policies via Conditional LoRA Generation**
  - A framework for *parameter-efficient adaptation* of imitation-learning policies, targeting *in-and-out distribution generalization* across changing object payloads and physical dynamics.
- **Zoom-Adaptive 3D Reconstruction and Localization via Online Camera Calibration**
  - A *unified framework* to adapt to changing zoom and non-linear camera models, enabling *zoom-aware Gaussian Splatting, robust 3D reconstruction, and accurate visual localization*.
- **Dynamic Object Manipulation for Humanoid Robots** *(Ongoing)*
  - A *learning-based visuomotor policies* for **dynamic object manipulation** on Unitree G1, focusing on real-time perception, motion prediction, and coordinated bimanual actions e.g. ball handling.

## Experience

**Research Associate** | **Turbomachines Research Lab** | ***IIT Kharagpur*** Aug 2020 -May 2022
Supervisor: Prof. Chetan S. Mistry

- Developed physics-based modeling and CFD simulation pipelines for compressor design and stability analysis.

**Research Intern** | **Turbomachines Research Lab** | ***IIT Kharagpur*** May 2019 - July 2019
Supervisor: Prof. Chetan S. Mistry

- Designed and evaluated a subsonic mixed-flow compressor using analytical modeling and CFD-based performance optimization.

**Research Intern** | **Turbomachines Research Lab** | ***IIT Kharagpur*** May 2018 - July 2018
Supervisor: Prof. Chetan S. Mistry

- Built theoretical and numerical models for supercritical $CO_2$ centrifugal compressors and validated simulations against experimental data.

**Research Intern** | **DRDO** | ***Chandigarh, India*** May 2017 - July 2017

- Modeled and simulated PDE engine dynamics using numerical methods and 3D CFD simulations

## Technical Skills

**Programming:** Python, C/C++ (familiar), MATLAB, PyTorch, OpenCV, Git, Linux, Docker.
**Robotics:** Imitation Learning, Diffusion Policies, RL, VLAs, Multimodal Learning, Online Adaption, Calibration, SLAM, 3D Reconstruction, Localization, Navigation, Manipulation, Adaptive Control, Teleoperation.
**Robotics & Simulation:** ROS/ROS 2, NVIDIA Isaac Sim, MuJoCo, Gazebo.
**Hardware:** Franka Emika, Universal Robots, Neuromeka Indy7, SO-101, RealSense, Quest 3, Arduino.

## Past Publications/Projects

- **Hemant Kumar**, Chetan S. Mistry, *"Tip Clearance Mechanism in Mixed Flow Compressor,"* Propulsion and Power Research, 2023. [Link]
- **Hemant Kumar**, Chetan S. Mistry, *"Implementation of Similarity Principle and Scaling Laws for Low-to-High Speed Mixed Flow Compressor,"* ASME Turbo Expo, Boston, 2023. [Link]
- **Hemant Kumar**, Chetan S. Mistry, *"Numerical Investigations on Aerodynamic Design Criteria for Low Speed Mixed Flow Compressor,"* ASME GT India, 2021. [Link]
- **Hemant Kumar**, Chetan S. Mistry, *"Numerical Performance and Flow Field Study of Centrifugal Compressor with Supercritical Carbon Dioxide,"* ASME GT India, 2019. [Link]
- **Advanced Machine Learning: Theory and Applications**, University of Maryland (Feb–June 2023) - Developed deep neural and reinforcement learning algorithms for autonomous course navigation.
- **Equilibrium Programming**, University of Maryland (Feb–June 2023) - Implemented a bi-level optimization framework using game-theoretic principles for energy market.

- **AIAA Design Competition 2019** - Ranked among the **Top 8 globally** for hybrid-electric UAV engine design, organized by the American Institute of Aeronautics and Astronautics.

## Academic Service

**Reviewer** : IROS, ICRA, RA-L, AAAI, ICRA'25, CPHS'26, Robotics and Computer-Integrated Manufacturing, ASME.
**Graduate Teaching Assistant** :ENME202: Computing Fundamentals for Eng, UMD, College Park. *Fall 2026*.
**Graduate Teaching Assistant** : ENME202: Computing Fundamentals for Eng, UMD, College Park. *Spring 2026*.
**Graduate Teaching Assistant** : ENES221: Dynamics, UMD, College Park. *Spring 2025*.
**Graduate Teaching Assistant** : ENES102: Mechanics-I, UMD, College Park. *Fall'22, Spring'23, Fall'23*
**Graduate Teaching Assistant** : AE39003: Structure Lab-II, IIT, Kharagpur. *Fall 2020*.

## Leadership & Honors

- **Future Faculty Program Travel Award**, University of Maryland. *2026*
- **Invention of the Year Award, Finalist**, UMD'25, for RACE: Real-time Adaptive Camera Estimation. *2025*
- **Dean's Fellowship**, University of Maryland, College Park. *2023*
- Conferred the prestigious **Institute Blue Award** for outstanding contribution to Basketball, IIT Kharagpur. *2020*
- Recipient of the **AICTE Postgraduate Scholarship**, Government of India, for academic excellence. *2019-2020*
- **Captain**, IIT Kharagpur Men's Basketball Team - **Silver Medal**, $54^{th}$ Inter IIT Sports Meet. *2019–20*
- **Group Leader**, National Service Scheme (NSS) - led community service initiatives. *2016–17*